

# The Angenent Oval under Curve Shortening: Exact Ancient Geometry and Two Grim-Reaper Ends

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## Abstract

For the central component of  $\cos x = e^t \cosh y$  in  $|x| < \pi/2$ ,  $t < 0$ , we give a convention-explicit proof of curve shortening, smooth embedded strict convexity, and exact curvature extrema.

## 1 Exact ancient solution and curvature

We use the convention that a convex curve has positive curvature  $\kappa$  with respect to its outward normal  $n$ , and curve shortening has outward normal velocity  $V_n = -\kappa$ . Put  $r = e^t \in (0, 1)$  and

$$F(x, y, t) = \cos x - r \cosh y, \quad \Gamma_t = \{F = 0, |x| < \pi/2\}.$$

**Theorem 1** (Core implicit solution). *For every  $t < 0$ , the central component  $\Gamma_t$  is a smooth embedded strictly convex closed curve. It evolves by  $V_n = -\kappa$ , where*

$$\kappa(x, y, t) = \frac{\cos x}{\sqrt{1 - e^{2t}}}, \quad \frac{e^t}{\sqrt{1 - e^{2t}}} \leq \kappa \leq \frac{1}{\sqrt{1 - e^{2t}}}. \quad (1)$$

*It is ancient on  $(-\infty, 0)$  and contracts to the origin as  $t \uparrow 0$ .*

*Proof.* The central component consists of the two graphs

$$y = \pm \operatorname{arcosh}(r^{-1} \cos x), \quad |x| \leq \arccos r.$$

On this level,

$$|\nabla F|^2 = \sin^2 x + r^2 \sinh^2 y = 1 - r^2 > 0, \quad D^2 F = -\cos x I. \quad (2)$$

Thus the level is smooth and compact. For  $n = -\nabla F/|\nabla F|$ , the implicit curvature formula and (2) give (1), which is positive because  $r \leq \cos x \leq 1$ . Differentiating  $F = 0$  along a moving point yields  $V_n = F_t/|\nabla F| = -\cos x/\sqrt{1 - r^2} = -\kappa$ . The displayed graphs shrink to  $(0, 0)$  as  $r \uparrow 1$  and exist for every  $r > 0$ , proving the temporal statements and the sharp extrema. The unrestricted zero set of  $F$  is the disjoint union  $\bigsqcup_{k \in \mathbb{Z}} (\Gamma_t + (2\pi k, 0))$ ; it is not one compact curve.  $\square$

## Source lineage

## References

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- [3] T. Bourni, M. Langford, and G. Tinaglia, “Convex ancient solutions to curve shortening flow,” arXiv:[1903.02022](https://arxiv.org/abs/1903.02022).